

Parallel Robots

EPFL – 2026

Arda YIĞIT



Statics

- Consider robot kinematics

$$\dot{\mathbf{q}} = \mathbf{J}^{-1} \dot{\mathbf{x}}$$

- Virtual work principle: a mechanical system is in equilibrium when the total virtual work done by all forces for any arbitrary infinitesimal virtual displacement is zero

- We can show that

$$\mathbf{J}^{-T} \boldsymbol{\tau} = \mathbf{w}$$

with $\boldsymbol{\tau}$ generalized actuation forces and $\mathbf{w}$ the output wrench

- For robots that are **not underactuated**, the statics only depend on the geometry

- For a non-redundant robot, let's find joint forces to compensate for gravity

- For a non-redundant robot pushing towards the wall, let's find actuation forces

- Statics model is useful for simplified dynamics models
- Example: only consider inertia at the actuators and/or end-effector

- Also sometimes called “kinematostatics”
- Consider that, rather than knowing joint and/or task coordinates, we measure actuation and external forces
 - Are there such cases?
- Then, joint and task coordinates can be found by solving

$$\begin{bmatrix} \mathbf{f}_{imp}(\mathbf{x}, \mathbf{q}) \\ \mathbf{J}^{-T} \boldsymbol{\tau} = \mathbf{w} \end{bmatrix} = 0$$

- We can have an equivalent method based on potential energy

- Let U be the total potential energy of the robot

$$U(\mathbf{q}, \mathbf{x}) = -W_{act}(\mathbf{q}) - W_{ext}(\mathbf{x})$$

- W_{act} and W_{ext} are respectively actuator work and external work

$$W_{act}(\mathbf{q}) = \boldsymbol{\tau}^T \mathbf{q}$$

$$W_{ext}(\mathbf{x}) = \mathbf{f}_{ext}^T \mathbf{q}$$

- Constrained optimization problem

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{q}} U(\mathbf{q}, \mathbf{x}) \\ \text{s.t. } \mathbf{f}_{imp}(\mathbf{q}, \mathbf{x}) = \mathbf{0} \end{aligned}$$

- First-order optimality condition

$$\nabla \mathcal{L}(\mathbf{q}, \mathbf{x}, \boldsymbol{\lambda}) = \mathbf{0}$$

with $\mathcal{L}$ the Lagrangian

$$\mathcal{L} = U + \boldsymbol{\lambda}^T \mathbf{f}_{imp}$$

- Limit case: non-conservative forces